

WASP-12b

R-0351

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231127 · created 2026-07-18T09:59:05+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460274.947 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.044
Transit depth (Rp/Rs)^2	2.35 +/- 1.36 [%]
Orbital Inclination (inc)	85.71 +/- 2.83
Airmass coefficient 1 (a1)	1.47 +/- 0.84
Airmass coefficient 2 (a2)	-0.3 +/- 0.84
Scatter in the residuals of the lightcurve fit is	57.25 %
Best Comparison Star	None
Optimal Aperture	4.12
Optimal Annulus	8.54
Transit Duration (day)	0.129 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.044 vs 0.1178 (deviation 0.8σ)
Duration fit vs published	0.129 d vs 0.125 d (deviation 0.85σ)
Mid-time O–C	-1569.8 min
Depth SNR	3.5
Residual scatter	57.25 %

Light curve

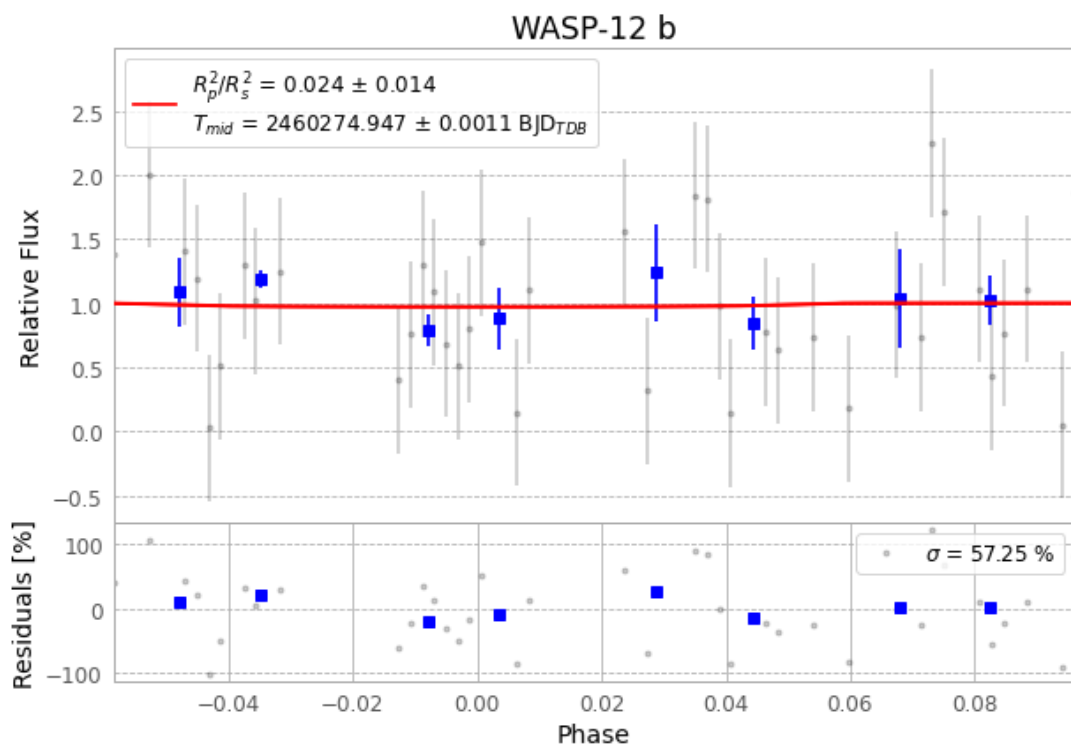


Figure 1 — FinalLightCurve_WASP-12 b_2023-11-26.png (SHA-256 0482d6fc8e35fccf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 360], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 22c55e1841763a94...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

83 FITS frames (55 MB), WASP-12231126090015.FITS → WASP-12231126130615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231127.log (d4cc796a260b...), FinalLightCurve_WASP-12 b_2023-11-26.png (0482d6fc8e35...), FinalLightCurve_WASP-12 b_2023-11-26.csv (99a052528c71...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 09:59Z.